Solenoid Valves Manual

Kaveri Motion Components · Rev B, Nov 2024

Contents and symbols

This manual covers installation, operation and maintenance. Read the Safety Instructions before working on the product. Symbols: WARNING marks a hazard that can cause injury; NOTE marks information that prevents damage.

Technical data

Function 5/2-way, single solenoid with spring return (SV52-S) or double solenoid (SV52-D). Pilot operated with internal pilot supply. Port size G1/4. Operating pressure 1.5 to 8 bar. Minimum pilot pressure is 1.5 bar; below 1.5 bar the valve will not switch reliably. For pressures below 1.5 bar or vacuum use the external pilot version SV52-EP (non-stock, 3 week lead time). Nominal flow 1100 Nl/min. Response time ON 18 ms, OFF 25 ms. Medium filtered air 40 micron. Ambient temperature -10 to +50 deg C.

Coils and electrical data

Coil 24 V DC, 3.0 W (code -24D, spare coil SV-COIL-24D). Coil 230 V AC 50/60 Hz, 4.5 VA holding (code -230A, spare coil SV-COIL-230A). Voltage tolerance +/-10 percent. Duty cycle 100 percent ED. Protection IP65 with the supplied connector to DIN EN 175301-803 form B. Never energise a coil that is removed from the armature tube; it will burn out. Ordering example SV52-S-24D: single solenoid, 24 V DC.

Manifold mounting

SV52 valves can be mounted individually or on manifolds of 2 to 10 stations. Refer to the manifold data sheet.

Troubleshooting - valve does not switch

If the valve does not switch when energised: 1) check that supply pressure at port 1 is at least 1.5 bar; 2) measure coil voltage at the connector, it must be within +/-10 percent of rated voltage; 3) operate the manual override. If the valve switches with the manual override but not electrically, the coil or wiring is faulty. If it does not switch with the override either, suspect low pressure or contamination. Continuous leakage from the exhaust port indicates worn spool seals or contamination; return the valve for service.